

Binary Social Game: A Principle for Enhancing Desired Behaviors through Asynchronous Social Interactions

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ABSTRACT

Within the HCI community, the impact of social dynamics on behavior change has gained considerable interest. This paper introduces the Binary Social Game (BSG), a principle that utilizes asynchronous social interactions to motivate desired behaviors. Initially, we developed GrowFlower based on the BSG principle, and its effectiveness was evaluated through a two-week wizard-of-oz study. After gaining insights from the preliminary study, we refined our approach and created GENGO, a bingo-style game that is specifically designed to encourage exercise as a targeted behavior. A subsequent two-month user study in real-world conditions, involving 82 participants, validated GENGO's effectiveness in fostering consistent exercise habits. These findings highlight the potential of BSG as a useful tool for promoting desired behaviors within a group context, especially through asynchronous social interactions. This study contributes to the understanding of how social dynamics can be leveraged for behavior change.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative and social computing**.

KEYWORDS

group game, behavior change, exercise motivation, social effects

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1 INTRODUCTION

Over the years, the human-computer interaction (HCI) community has shown a keen interest in gamification, as they foster motivation, engagement, and offer potential behavior change benefits [7, 20]. One of key elements to motivate users to induce target behaviors is social effects (or social interaction) [1, 4, 12]. In particular, social effects among group members includes creating a group to share members' status, collaboration to achieve mutual goals, and competition between groups [8, 19].

Social interactions can serve as a effective motivator for target behavior, whether in real-time [2, 5, 10, 16] or asynchronous interactions [1, 15, 18]. Asynchronous interactions, in particular, offer the unique advantage of time flexibility, enabling individuals to engage in the desired behavior at their convenience. While this flexibility may increase the probability of successful behavior change, potentially leading to improved outcomes [3, 13], proper triggering is also an important factor to perform the behavior [9].

In this paper, we introduce a novel design principle—Binary Social Games (BSG)—aimed at facilitating desired behaviors through asynchronous interaction in a group setting. This approach incorporates three features including binary outcomes of target behaviors, explicit turn-taking for nudging, and the strong influence of negative reinforcement through social consequences. The BSG model seeks to engage group members, encouraging them towards achieving desired behaviors by capitalizing on these social elements.

To assess the feasibility of this concept, we designed GrowFlower, a game grounded in the BSG principle, and conducted a two-week wizard-of-oz study involving eight participants. Based on the insights gained from the preliminary study, we iteratively refined GrowFlower, developed it into GENGO – a bingo-style game specifically designed to promote exercise within group settings. To validate its effectiveness in real-world settings, we developed a fully functional mobile application and conducted a two-month user study with 82 users. The results showed the potential of the BSG concept, as the participants reported improved exercise habits due to GENGO. The implications of our findings hold considerable promise for the HCI community, contributing to the burgeoning body of research on utilizing social dynamics to influence behavior.

2 CONCEPT OF BINARY SOCIAL GAME

We were primarily inspired by two papers, Step-by-Step [17] and BeUpright [19]. Step-by-Step demonstrated that explicitly assigning a turn to exercise by passing a box to the next co-worker effectively initiated physical movements. BeUpright proposed the Relational Norm Intervention model, which leverages the psychology of individuals who strive to avoid making others uncomfortable.

Building upon the insights gained from previous studies, we propose the concept of the *Binary Social Game*, which incorporates these findings into a social game driven by asynchronous interaction among group members, utilizing only binary results of the target behavior. The utilization of binary results simplifies implementation, as the system solely focuses on detecting whether the target behavior occurs. The overall mechanism of the Binary Social Game consists of three components:

- ① **Binary result of target behavior:** The system utilizes the outcomes indicating whether group members performed the target behavior (e.g., completion of daily exercise, drinking water, or studying) as input.
- ② **Explicit turn to do for nudging:** The system explicitly assigns turns to individual members for performing target behavior. This explicit turn allocation also creates implicit social pressure, as other members await the completion of the task.
- ③ **Negative reinforcement through social effects:** When a member fails to fulfill the target behavior within a specific time period, the entire group incurs penalties.

3 PRELIMINARY STUDY

3.1 GrowFlower Design

Based on the design considerations of the Binary Social Game (BSG), we developed *GrowFlower* (the concept is illustrated in Figure 1), a simplified social game in which team members collaborate to cultivate a flower, aiming to motivate individuals to engage in daily exercise. Each member is required to exercise for at least 10 minutes once a day (① *Binary result of target behavior*) in order to water the flower when it is their turn (② *Explicit turn to do for nudging*). If all members successfully water the flower (complete their exercise for the day), it thrives and continues to grow. However, if any member fails to water the flower (does not complete their exercise for the day), it withers and dies (③ *Negative reinforcement through social effects*), prompting the group to restart the process. The sequence of exercise is randomly assigned each day.

3.2 Wizard of Oz Study

We employed a Wizard of Oz approach to evaluate the feasibility of *GrowFlower*. The study was conducted over a period of 2 weeks and involved 8 participants (7 male, 1 female). All participants were office workers aged between 25 and 40 years. Two participants had an active lifestyle, engaging in exercise 2–3 times a week, while the others did not have a regular exercise routine. Each group consisted of three inactive participants and one physically active participant, resulting in a total of 2 groups. The assignment of participants to groups was randomized.

For the study infrastructure, we utilized KakaoTalk, the most popular mobile messenger in South Korea. A group chat was created

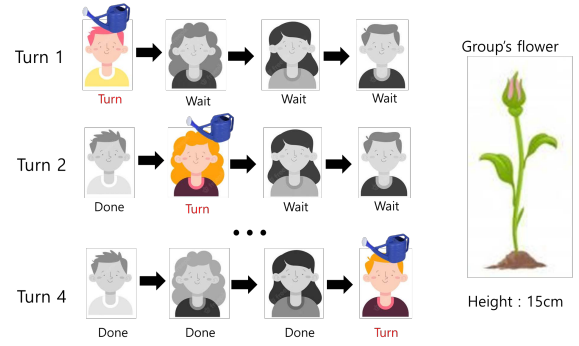


Figure 1: The overall concept of GrowFlower.

to facilitate communication among the 8 participants and a moderator. Every morning, the moderator provided updates on the latest flower status and posted randomly assigned exercise sequences for both groups. Participants were instructed to exercise for at least 10 minutes each day and were required to upload their exercise details and heart-rate data, measured by a smartwatch. Whenever a participant completed their exercise and uploaded the results, the moderator updated the group chat with the progress of the group exercise. Each group lasted for one week, after which the members were randomly reassigned for the following week.

Following the 2-week study period, a focus-group interview was conducted with all participants. The interview primarily focused on questions such as "How did you feel when it was your turn to exercise?", "What did you do if other members were not exercising?", and "Does it help motivate you to exercise consistently?". The interview session lasted for approximately 30 minutes.

3.3 Preliminary Findings

The interview revealed that participants felt uneasy when it was their turn to water the flower and had to exercise because other members were waiting for them. This demonstrates that giving an explicit turn to exercise is an effective nudge for promoting target behavior. Additionally, it triggered social interaction among the members, primarily in the form of participants waiting "in line" and encouraging others to exercise as early as possible. This finding is noteworthy as it indicates that members engaged in supportive interactions [8].

On the other hand, participants who exercised frequently expressed dissatisfaction since they were unable to exercise until other participants completed their daily exercise. Despite having fixed exercise times, they had to wait if the member in front of them did not exercise. Conversely, most participants who rarely exercised acknowledged the positive effect of the sequential approach for behavior change. This result suggests that the combination of members' characteristics may influence overall group satisfaction in BSG design.

In summary, we identified both pros and cons associated with the rule of compulsory sequential turn. The advantage was that it provided stronger reinforcement for the person in the exercise order, while the downside was that individuals at the back of the

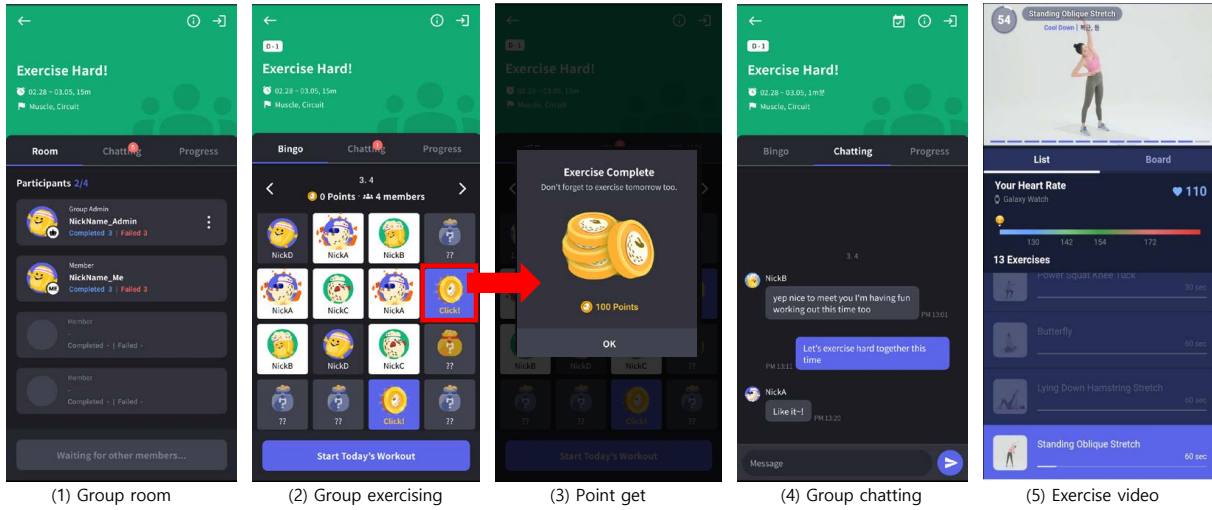


Figure 2: The example screenshots of GENGO

order had to wait until their turn arrived, potentially causing them to miss out on exercise due to their busy real-life schedules. To mitigate this issue, bingo-style rules were implemented in GENGO to alleviate the compulsion of the exercise sequence and facilitate asynchronous interactions.

4 GENGO MOBILE APPLICATION DESIGN

We propose GENGO¹, a bingo-style game design that follows the concept of BSG for individuals to motivate for regular exercise. In GENGO, when a member completes an exercise, the corresponding bingo pieces assigned to that member are turned over (① *Binary result of target behavior*). When any line of the bingo board is fully turned over, the group earns points (② *Explicit turn to do for nudging* and ③ *Negative reinforcement through social effects*). Also, when group activities happen, like exercise completion, push notifications are sent to all members. The overall features and process of GENGO are shown in Figure 2 and summarized below:

- (1) **Group room:** Any user can create a group with desired exercise style. The number of group members can range from 2 to 4. The more members present in a group, the more difficult it becomes to complete a bingo line, so there is a higher chance of earning more points in bingo. A room admin can start the group for exercising. In the group room, members can check each other's history of completed and failed previous group exercises.
- (2) **Group exercising:** A new bingo board is created every day at 12:00 AM. There are 9 bingo pieces, which are randomly assigned to each member. For example, in a group of 2 members, each member receives 4-5 pieces, while in a group of 4 members, each member receives 2-3 pieces. When a particular member completes an exercise, all their assigned bingo pieces are turned over, and the others receive push notifications. Upon pressing

the start button, a 10-minute body-weight home-training video² is played (shown in Figure 2 (5)). Once the video ends, GENGO recognizes that the member's exercise is complete.

- (3) **Point acquisition:** Whenever a line is completed, points can be obtained by tapping on the pocket at the edge. In a day when all members complete the exercise, up to 7 pockets can be obtained.
- (4) **Group chatting:** Members can encourage and support each other through the chat feature.

5 METHODOLOGY

We conducted a two-month user study in the wild to evaluate the effects of regular exercise, aiming to minimize any external compulsion that may arise in an experimental setting. Conducting studies in the wild is known to be challenging [14], but it provides a reliable method for capturing the real-world usage of a new service [6, 11].

For the field test, we sent a beta service email to the employees in our company, along with a download URL. Over two hundred users registered for our service, but only 82 participants (11 in their 20s, including 6 males; 35 in their 30s, including 25 males; and 36 in their 40s, including 32 males) joined a group and completed at least one exercise session. We did not provide compensation to every participant, but we informed them that the top three ranked groups would receive a gift of \$100 per person in the group. Each group can consist of 2 to 4 members and lasts for one week. Each member is required to complete 10 minutes of exercises with our service per day, for a total of seven times within a week. The group points awarded in the pocket were randomly calculated based on the number of group members because the more group members, the more difficult they are. Furthermore, groups were ranked based on the points accumulated throughout the 7-day group exercise, which aims to induce negative reinforcement (i.e., if someone doesn't exercise, their group won't get points). Following the study, we sent

¹GENGO stands for Group Exercise through biNGO.

²The exercise videos were produced by hiring professional trainers in-house, and various exercise videos were supported according to the user's exercise level.

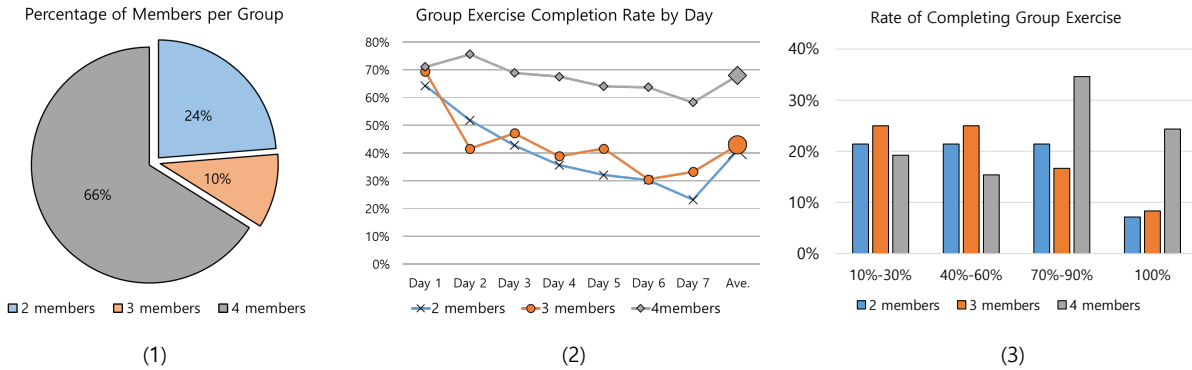


Figure 3: (1) Percentage of members per group in the field test, (2) Average of group exercise completion rate by day (100% when every member exercised in a group), and (3) Rate of completing group exercise during 7-days

emails to actively participating users, inviting them to participate in user interviews. Finally, we conducted one-on-one interviews with five participants, discussing their experiences using the application through online meetings. Each interview lasted for 30 minutes, and participants received compensation of approximately \$10.

6 RESULTS

During the two-month user study conducted in the wild, a total of 118 groups successfully gathered members, initiated activities, and completed at least one exercise (some groups failed to start due to insufficient members, absence of a room admin in the application, etc.). The usage patterns of these groups are illustrated in Figure 3.

Effectiveness of GENGO: Most of the participants stated that the bingo-style group game motivated them to complete their daily exercise. One participant said, "I would feel very sorry if other team members couldn't get points because of me. I exercised to avoid harming my teammates" (P2). Another participant said, "When I received push-notifications that everyone else in my group finished their workout except for me, I felt the pressure to do it quickly" (P5). Additionally, some participants encouraged each other to complete a bingo line, as expressed by one participant: "I encouraged my teammates through chat to complete the bingo. I didn't directly tell them to exercise, but I wrote 'cheer up!'" (P1). However, one participant mentioned a negative experience: "I was full of willpower to work hard, but there was a time when a team member didn't exercise consistently, so I left the group and formed a new one. It was quite a depressing experience. After that, I started filtering my group members" (P4).

Importance of Group Dynamics: The majority of users preferred to join 4-member groups (66%, 78 out of 118). Over time, as the duration of group activities progressed, the group exercise completion rate (number of members who completed exercise / number of group members) decreased. However, among the different group sizes, the 4-member groups exhibited the smallest decline. Notably, participants maintained a higher average completion rate of exercise sessions per week which increased with the number of group members. Specifically, groups with two members achieved a 40% completion rate, exercising 2.8 times per week on average, groups

with three members had a 43% completion rate, exercising three times per week on average, and groups with four members notably achieved a 68% completion rate, exercising 4.8 times per week on average. Furthermore, the ratio of groups that achieved 100% exercise completion was significantly higher among these groups, in comparison to those with other group sizes.

This finding may be attributed to two reasons: (1) the expected points increase in groups with more members, and (2) the potentially more powerful social effect of negative reinforcement in larger groups. Furthermore, we observed that individuals who regularly engaged in exercise predominantly participated in 4-member groups and frequently formed repeated groups among themselves, suggesting the importance of coordination among group members.

Also, so there were groups comprised of members who already knew each other, as well as cases where members who were previously strangers formed groups. All participants agreed that group solidarity influenced the enjoyment and participation in exercise, and one participant noted that a sense of solidarity even developed within groups of strangers: "I started with people I didn't know at the beginning, but afterward, we continued to exercise by creating a room with the same people. It felt like I made an online friend who works out" (P1). However, one participant expressed that when strangers formed groups, the group tended to disband prematurely: "I feel that forming a group with people I don't know doesn't create a sense of solidarity. There were often cases where group activities were stopped due to someone suddenly leaving the group" (P2).

7 FUTURE WORK

Future work could involve conducting a more in-depth exploration of various group dynamics. While our study focused primarily on groups ranging from two to four members, further investigation into factors such as group size and composition (homogeneous vs. diverse, pre-acquainted vs. strangers) could offer valuable insights into optimal configurations for Binary Social Game (BSG). Additionally, it would be worthwhile to explore the applicability of BSGs in other behavior change domains, such as diet, study habits, or medication adherence, to expand the versatility of BSGs.

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